

Social Experience Shapes Fighting Strategies for Reproductive Success

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The paper presents a new behavioral assay for *Drosophila* aggression and demonstrates that social experience influences fighting strategies, with group-housed males favoring high-intensity but low-frequency tussling over aggressive lunging observed in isolated males. This paper is **valuable** for researchers studying *Drosophila* social behaviors, while the characterization of behavioral and neuroanatomical data is **incomplete**.

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Abstract

Social isolation generally increases aggression but decreases mating competition, resulting in an intricate and ambiguous relationship between social experience, aggression, and reproductive success. In male *Drosophila*, aggression is often characterized by lunging, a frequent and comparatively gentle combat behavior. Here, we establish a behavioral paradigm for studying a less frequent but more vigorous fighting form known as tussling. We discover that while social enrichment decreases lunging, aligning with past observations, it heightens the more forceful tussling behavior. These two forms of aggression rely on different olfactory receptor neurons, specifically Or67d for lunging and Or47b for tussling. We further identify three pairs of central pC1 neurons that specifically promote tussling. Moreover, shifting from lunging to tussling in socially enriched males facilitates better territory control and mating success, mitigating the disadvantages associated with aging. These findings reveal how social experience shapes fighting strategies to optimize reproductive success.

Introduction

Competition for resources such as territory, food, and mates is crucial for animal survival and reproduction [1–4]. Among diverse competitive strategies, physical combat is a prevalent form of fighting to repel rivals and regulated by both genetic factors and social environment [1, 5].

5[2]. Understanding how social experience influences combat behaviors in animals to maximize their survival and reproduction remains a fundamentally important yet inadequately understood question.

In the fruit fly, *Drosophila melanogaster*, inter-male aggression includes low-intensity fights such as lunging, in which one fly rears up on its hind legs and snaps down onto its opponent, as well as high-intensity fights like tussling, where two males tumble over each other [6[2]-8[2]]. It is noteworthy that the majority of studies focus on the low-intensity lunging behavior, owing to its higher frequency and ease of observation. Substantial studies have identified regulatory genes and the neural circuit underlying the lunging-based aggressive behaviors [7[2], 9[2]]. In particular, the male-specific volatile pheromone 11-*cis*-vaccenyl acetate (cVA) acutely promotes lunging behavior via its olfactory receptor Or67d [10[2]]. Interestingly, chronic cVA exposure in socially-enriched males reduces the level of lunging via another olfactory receptor Or65a [11[2]]. Consistently, it has been found that social isolation increases lunging frequency by altering the expression of genes such as *Cyp6a20* [12[2]], *Drosulfakinin (Dsk)* [13[2]], and *Tachykinin (Tk)* [14[2]]. Social isolation also increases male aggression in mice [15[2], 16[2]] and even in humans [17[2]]. In mice, the TK homologue Tachykinin 2 (Tac2) responds to social isolation and promotes inter-male aggressive behaviors [16[2]]. These findings contribute to the consensus that social isolation escalates aggression, whereas social enrichment mitigates it in animal models.

Meanwhile, social enrichment increases male mating success by enhancing the sensitivity of *Or47b*-expressing olfactory receptor neurons (ORNs) to fly pheromones common to both sexes [18[2], 19[2]]. Artificially activating *Or47b* ORNs also improves mating advantage in a competitive assay [20[2]]. Group housing and age-dependent juvenile hormone (JH) jointly increase the expression of the male-specific proteins from *fruitless (fru^M)*, which is a master gene for male courtship behaviors [21[2]-23[2]], in the *Or47b*-expressing ORNs [19[2], 24[2], 25[2]]. Notably, courtship levels in single-housed and group-housed wild-type males are not significantly different [26[2]]. In fact, group-housing decreases excitability of the courtship-promoting P1 neurons [27[2]], which express both terminal genes in the sex determination hierarchy, *fru* and *doublesex (dsx)*. These findings raise a question on how social enrichment, which suppresses lunging-based aggression and does not increase courtship intensity, would facilitate mating competition.

Here, we establish a behavioral paradigm for studying inter-male tussling in *Drosophila* and discover that this infrequent yet intense form of combat is enhanced by social enrichment and is crucial for territory dominance and mating competition. This process requires olfactory receptor Or47b, but not Or67d or Or65a. We further identify three pairs of central *dsx*-expressing pC1 neurons that specifically promote tussling. Moreover, we demonstrate that socially enriched males can overcome mating disadvantages associated with aging.

Results

A Paradigm for High-intensity Aggression in *Drosophila*

While studying the lunging-based aggression in *Drosophila*, we occasionally observed the high-intensity tussling behavior in male flies. We reasoned that the intensity, instead of the frequency, could be decisive for a combat. Thus, although tussling is a less frequent behavior, it could serve a vital function.

To induce a higher level of tussling for further manipulation, we first introduced a fixed live virgin female in the same behavioral chamber for assaying lunging. The female fly was immobilized in the middle of the food, with its head embedded in the food and its body exposed to enhance male competition (Figure 1A[2]). We found that males that were group-housed for 7 days showed a significantly higher frequency of tussling behavior and reduced latency to tussle in the presence of

an embedded virgin female (**Figure 1B**). We next compared tussling in male flies of different ages. We observed a significantly higher proportion of 14-day-old males raised in groups that showed tussling behavior during the 2-hour observation period, compared to the 7-day-old males (**Figure 1B**). Furthermore, 2-day-old males never displayed tussling behavior. In contrast, 14-day-old males showed low level of lunging behavior than 7-day-old males did (**Figure 1**—figure supplement 1A and 1B). Previous studies have shown that, in addition to mating resources, food quality is another crucial factor affecting aggressive behaviors [28]. Therefore, we next assayed tussling behavior in 14-day-old males in the presence of an embedded virgin female with different concentrations of sucrose or yeast in the food during the behavioral test. We found that these males displayed similar levels of tussling behavior with different concentrations of sucrose in the food (**Figure 1C**). On the contrary, male tussling intensity increased significantly with higher yeast concentrations such that more than 70% of males displayed this high-intensity fighting behavior (**Figure 1D**, see Video 1). Taken together, these results establish an efficient behavioral paradigm for the high-intensity tussling behavior and indicate that older males display more tussling but less lunging than younger ones.

Social Enrichment Inhibits Low-intensity Lunging but Enhances High-intensity Tussling

Previous studies have shown that social enrichment inhibits aggressive behaviors, specifically the low-intensity lunging, in male flies. Therefore, we wondered whether social enrichment might also inhibit the high-intensity tussling behavior in males. To answer this question, we compared both fighting behaviors in single-housed (SH) and group-housed (GH) 14-day-old males (**Figure 2A**). Lunging and tussling events from each group of representative samples were exhibited in raster plots for comparison (**Figure 2B**). We scored lunging behavior in the first 10-min of test, due to its relatively higher occurrence, and tussling behavior during the whole 2-hour observational period since males take much longer time to initiate tussling (**Figure 1D**). Interestingly, we found that while SH males performed more frequent lunging behavior in 10 min than GH males did, which is consistent with previous findings, they rarely showed tussling behavior. In contrast, GH males displayed a large number of tussling events within 2 hours. Note that a tussling event may last up to several minutes, indicative of its high intensity, in contrast to lunging events that were often transient (**Figure 2B**). Statistical analysis of tussling and lunging behaviors showed that GH males exhibited reduced tussling latency and enhanced tussling frequency, in addition to increased lunging latency and decreased lunging frequency, compared to SH males (**Figure 2C-F**). These results indicate that social enrichment inhibits low-intensity lunging but promotes high-intensity tussling in male flies.

Or47b ORNs Mediate Social Experience-induced Tussling Behavior

To further investigate the sensory basis underlying the opposite regulation of lunging and tussling behaviors by social experience, we tried to examine the role of cVA-sensitive neurons, as well as other *fru^M*-expressing ORNs, in mediating intermale aggression. Two cVA sensory neurons, *Or67d* and *Or65a* ORNs, play crucial roles in enhancing the lunging-based aggression in SH males, in which *Or67d* ORNs mediate acute cVA stimulation and promote male aggression [10], while *Or65a* ORNs mediate chronic cVA stimulation and inhibit male aggression in GH males [11]. In addition, the *fru^M*-expressing *Or47b* ORNs are sensitive to a class of pheromones common to both sexes, such as palmitoleic acid (PA), and their sensitivity is enhanced under group rearing conditions [19]. *Ir84a* ORNs, which are also *fru^M*-positive, are involved in the perception of mixed food odors and promote male courtship behavior [29]. We silenced *Or67d*, *Or65a*, *Or47b* or *Ir84a* ORNs by expressing the inwardly rectifying potassium channel *Kir2.1* and assayed lunging and tussling behaviors in 14-day-old males. We found that silencing *Or47b* ORNs, but not *Or67d*, *Or65a* or *Ir84a* ORNs, significantly decreased male tussling, compared to the control

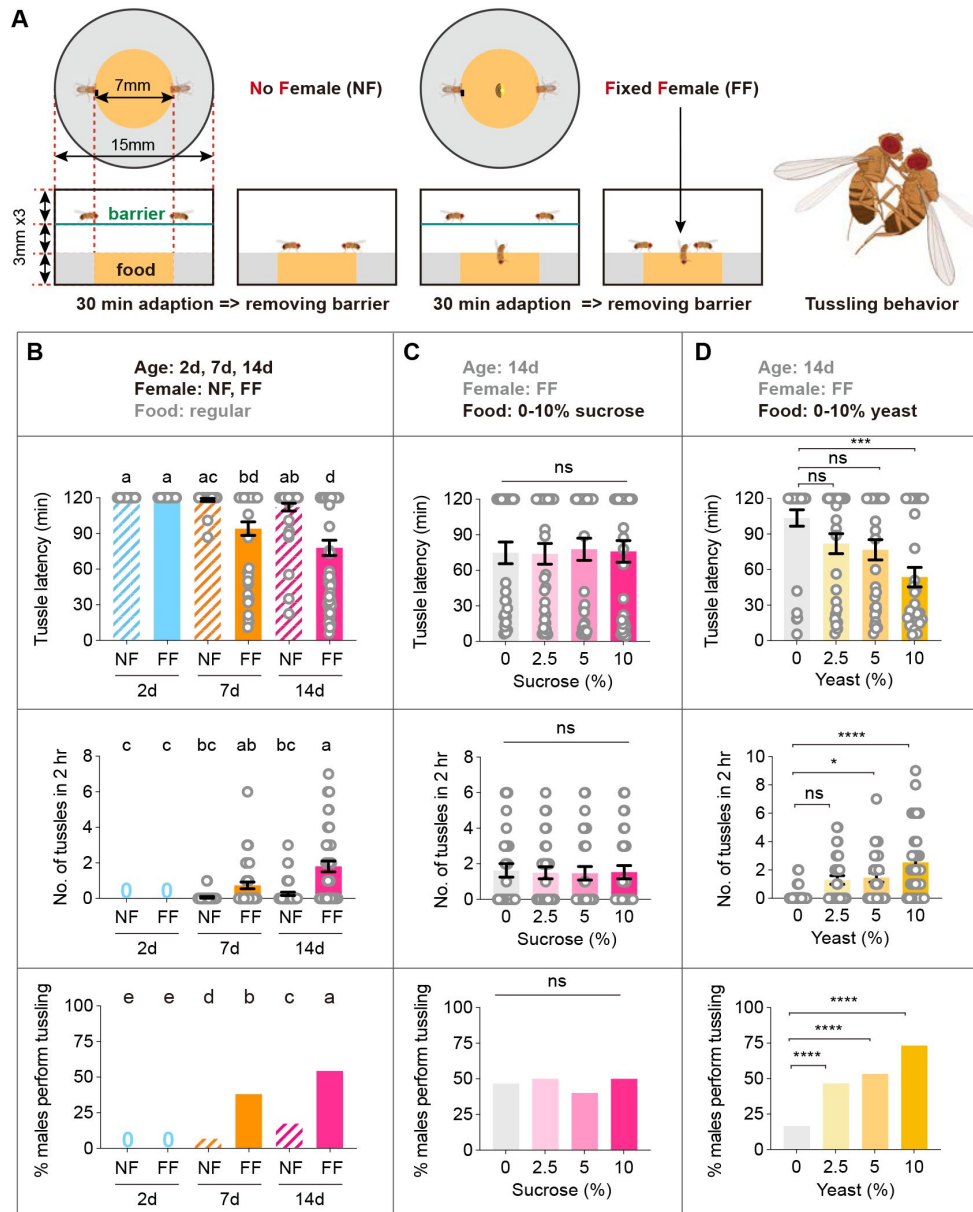


Figure 1.

Establishing a paradigm for tussling behavior in male *Drosophila*.

(A) Diagrams of two fighting paradigms for *Drosophila* tussling behavior. The left diagram shows a traditional fighting paradigm with no female (NF) in the food area and a new fighting paradigm with a fixed female (FF) in the center of the food area. Before the experiment began, two male flies were placed on the upper layer, separated from the food by a film. After a 30-minute adaptation, the film was removed to start recording. The right diagram shows a pair of tussling male *Drosophila*.

(B) The influence of age and female factors on male tussling behavior. From top to bottom, the tussle latency, the number of tussles, and the proportion of tussles occurring are shown respectively. $n=30, 30, 30, 46, 46, 46$ from left to right. Bars sharing the same letter are not significantly different according to Kruskal-Wallis test (tussle latency, number of tussles) and Chi-square test (proportion of tussles occurring).

(C) Sucrose concentration of food have no effect on male tussling behavior. $n=30$ for each group, n.s., not significant, Kruskal-Wallis test (tussle latency, number of tussles), and Chi-square test (proportion of tussles occurring).

(D) The influence of yeast concentration on male tussling behavior. $n=30$ for each group, n.s., not significant, * $P < 0.05$, *** $P < 0.001$, **** $P < 0.0001$, Kruskal-Wallis test (tussle latency, number of tussles), and Chi-square test (proportion of tussles occurring). Error bars correspond to SEM.

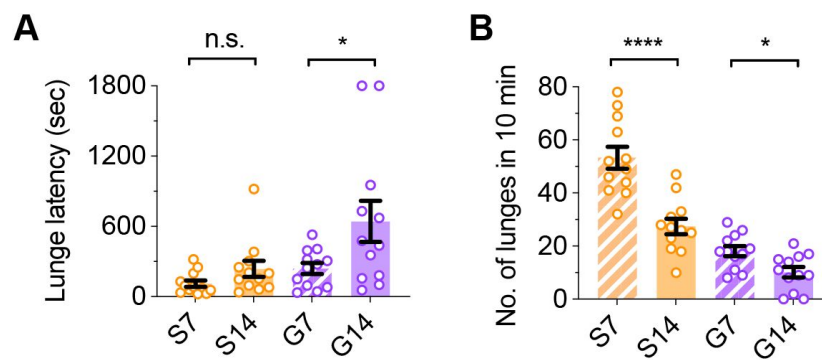


Figure 1—figure supplement 1.

Aging suppresses lunging behavior in male *Drosophila*.

(A) The lunge latency in S7, S14, G7 and G14 males, n=12 for each group, n.s., not significant, *P < 0.05, Mann-Whitney test. (B) The number of lunges in S7, S14, G7 and G14 males within 10 minutes, n=12 for each group, *P < 0.05, ****P < 0.0001, Mann-Whitney test. Error bars correspond to SEM.

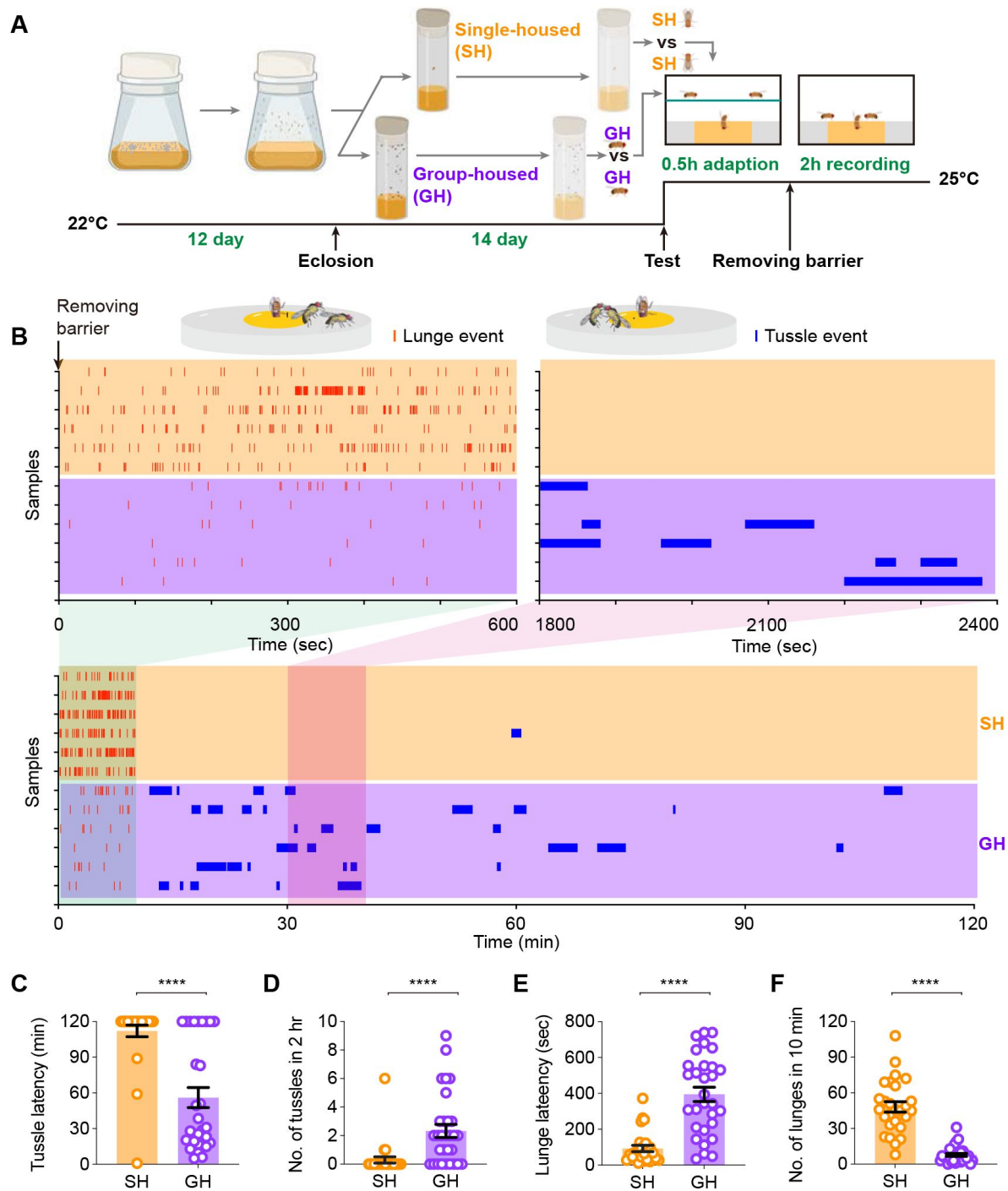


Figure 2.

Social enrichment inhibits lunging behavior but promotes tussling behavior in males.

(A) Schematic diagram of *Drosophila* rearing conditions and experimental procedure.

(B) Representative raster plots illustrating lunge events in the first 10 minutes and tussle events over 2 hours in single-housed (SH) and group-housed (GH) males.

(C, D) Latency (C) and number (D) of tussling behavior in SH and GH males, n=27, 30 for SH and GH males, respectively, ****P < 0.0001, Mann-Whitney test for both.

(E, F) Latency (E) and number (F) of lunging behavior in SH and GH males, n=27, 30 for SH and GH males, respectively, ****P < 0.0001, Mann-Whitney test for both. Error bars correspond to SEM.

(Figure 3A, B). In contrast, silencing *Or67d* ORNs, but not other ORNs, significantly reduced male lunging (Figure 3C, D). These loss-of-function results suggest that *Or47b* ORNs mediate male tussling behavior, whereas *Or67d* ORNs regulate male lunging behavior.

To elucidate the potential role of the *Or47b* receptor in mediating tussling behavior, we knocked down its expression in *Or47b* ORNs. Our results showed a significant decrease in male tussling behavior when the *Or47b* receptor expression was reduced (Figure 3E, F), indicating that the functionality of *Or47b* ORNs in tussling is dependent on the *Or47b* receptor. As previous studies suggested that the male-specific *fru^M* sustains *Or47b* expression in an activity-dependent manner [19], we next knocked down *fru^M* expression in *Or47b* ORNs and assayed male tussling. The efficiency of *Or47b* and *fru^M* RNAi knockdown was both validated by quantitative PCR (Figure 3—figure supplement 1A and 1B). We observed a slight increase of tussling latency in *fru^M* knocked-down males, although not significantly different from the control (Figure 3E, F), suggesting that *Or47b*, rather than *Fru^M*, plays a more crucial role in mediating male tussling.

To further confirm that the effect of social enrichment on tussling depends on the activity of *Or47b* ORNs, we artificially activated *Or47b* ORNs by expressing the bacterially derived sodium channel (NaChBac)[30] in 14-day-old SH males. We found that these SH males with enhanced activity of *Or47b* ORNs showed a significant increase in tussling behavior compared to control SH males. Interestingly, the level of tussling in these SH males with activated *Or47b* ORNs was indistinguishable from that in GH males (Figure 3G, H). These gain-of-function results indicate that social enrichment acts through *Or47b* ORNs to promote male tussling. Consistent with this, we observed substantially more downstream neurons of *Or47b* ORNs in GH males than those in SH males (Figure 3—figure supplement 2A and 2B), as visualized by the trans-Tango technique [31].

Distinct Central Brain Neurons for Lunging and Tussling

We next set out to identify the central neurons involved in regulating the high-intensity tussling. We found that silencing the mushroom body neurons via Kir2.1 did not affect male tussling, suggesting that the experience-dependent tussling behavior is not dependent on the classical learning and memory center. Surprisingly, inactivation of the previously-identified aggression-promoting P1^a and TK neurons did not affect tussling behavior (Figure 4A, B). These results imply that the central circuit for tussling behavior may be distinct from the previously reported aggression circuits. As *dsx*-expressing pC1 neurons, also termed P1 if co-expressing *fru^M*, are crucial for sexual and aggressive behaviors in both males and females, we also silenced two subsets of pC1 neurons (pC1^{SS1} and pC1^{SS2}), whose functions were previously characterized in females but not in males. We found that inactivation of pC1^{SS2} but not pC1^{SS1} neurons significantly reduced the tussling behavior in male flies (Figure 4A, B). The cell bodies of pC1^{SS2} and P1^a neurons are both located in the center of posterior brain and project to overlapping but distinct parts of the lateral protocerebral complex (Figure 4C-E). These results suggest that pC1^{SS2} neurons may be key central neurons regulating tussling behavior.

To further compare the function of P1^a and pC1^{SS2} neurons in regulating male behaviors, we performed optogenetic activation experiments on these neurons expressing *CsChrimson*. Compared to control male pairs that did not show any significant change upon red light stimulation (Figure 4F-H), optogenetic activation of P1^a neurons induced acute courtship behavior as indicated by unilateral wing extension (UWE), and upon red light removal, there was a significant increase in lunging but not tussling (Figure 4I-K), consistent with previous findings. In contrast, optogenetic activation of pC1^{SS2} neurons induced strong and acute tussling behavior, with more than 50% of the time during red light in tussling, and the tussling behavior ceased immediately upon removal of red light (Figure 4L, see Video 2). Activation of pC1^{SS2} neurons did not significantly affect either male lunging or courtship (Figure 4M, N). These results indicate that pC1^{SS2} neurons acutely promote the high-intensity tussling behavior.

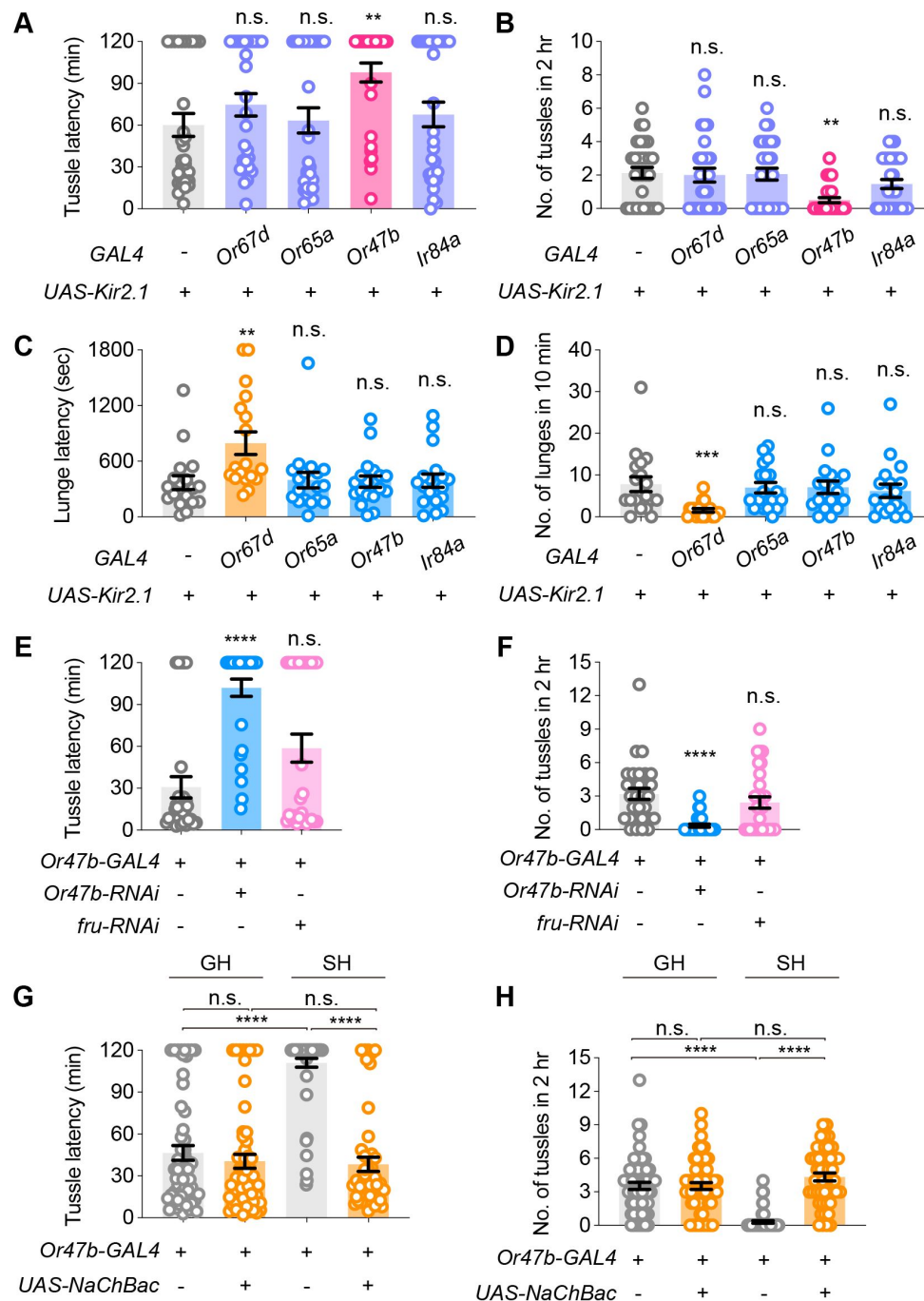


Figure 3.

Or47b neurons are required for male tussling but not lunging.

(A, B) The effect of inactivating different sensory neurons on latency (A) and number (B) of male tussling behavior, $n=30$ for each group, n.s., not significant, $**P < 0.01$, Kruskal-Wallis test.

(C, D) The effect of inactivating different sensory neurons on the latency (C) and number (D) of male lunging behavior, $n=30$ for each group, n.s., not significant, $**P < 0.01$, $***P < 0.001$, Kruskal-Wallis test.

(E, F) The effect of knocking down *Or47b* or *fru*^M expression in *Or47b* ORNs on the latency (E) and number (F) of male tussling behavior, respectively. $n=30$ for each group, n.s., not significant, $****P < 0.0001$, Kruskal-Wallis test.

(G, H) The effect of activating *Or47b* ORNs on latency (G) and number (H) of tussling behavior under conditions of GH or SH for 14 days, $n=60, 60, 60, 49$ respectively, n.s., not significant, $****P < 0.0001$, Kruskal-Wallis test. Error bars correspond to SEM.

Figure 3—figure supplement 1.

Validation of the *Or47b* and *fru*^M RNAi efficiency.

Validation of *UAS-Or47b-RNAi* (A) and *UAS-fru*^M (B) efficiency by quantitative qPCR. n=9 for each of three replicates, ****P < 0.0001, ***P < 0.001, Mann-Whitney test. Error bars correspond to SEM.

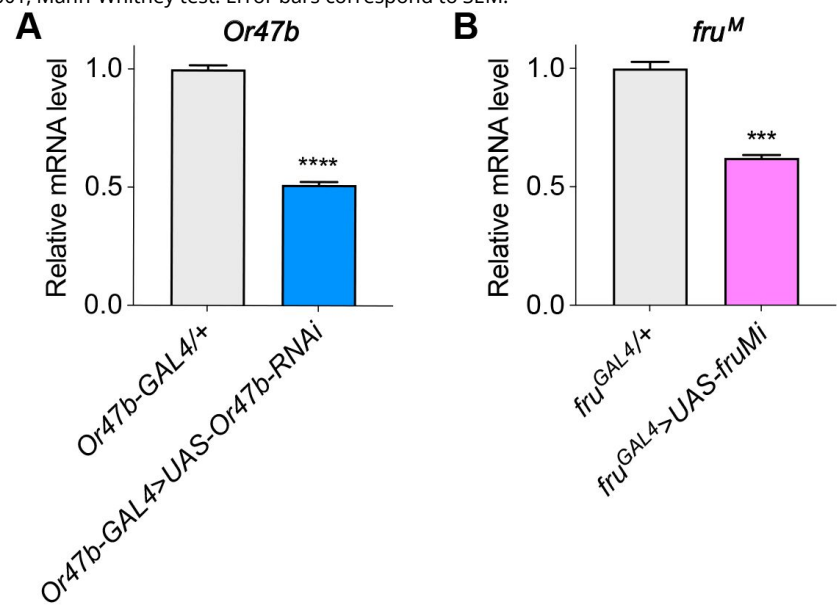
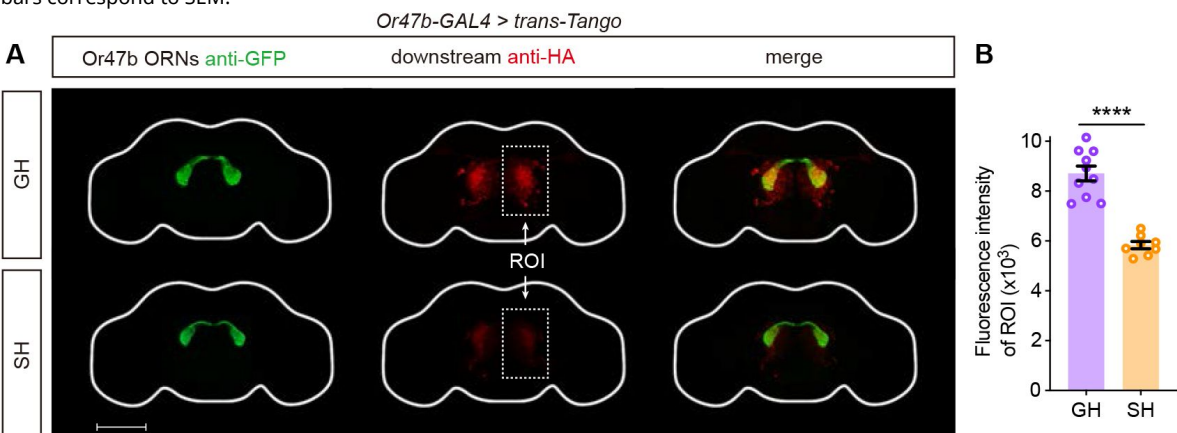


Figure 3—figure supplement 2.

Group housing increases downstream signals of *Or47b* ORNs.

(A) Projection of *Or47b* ORNs (green) and their downstream neurons (red) in GH and SH conditions. Scale bar, 100µm. (B) The fluorescence intensity of neurons downstream of *Or47b* ORNs is stronger under GH conditions compared to SH conditions. Both groups of male testers were 14 days old, n=10 and 8, respectively, ****P < 0.0001, Mann-Whitney test. Error bars correspond to SEM.



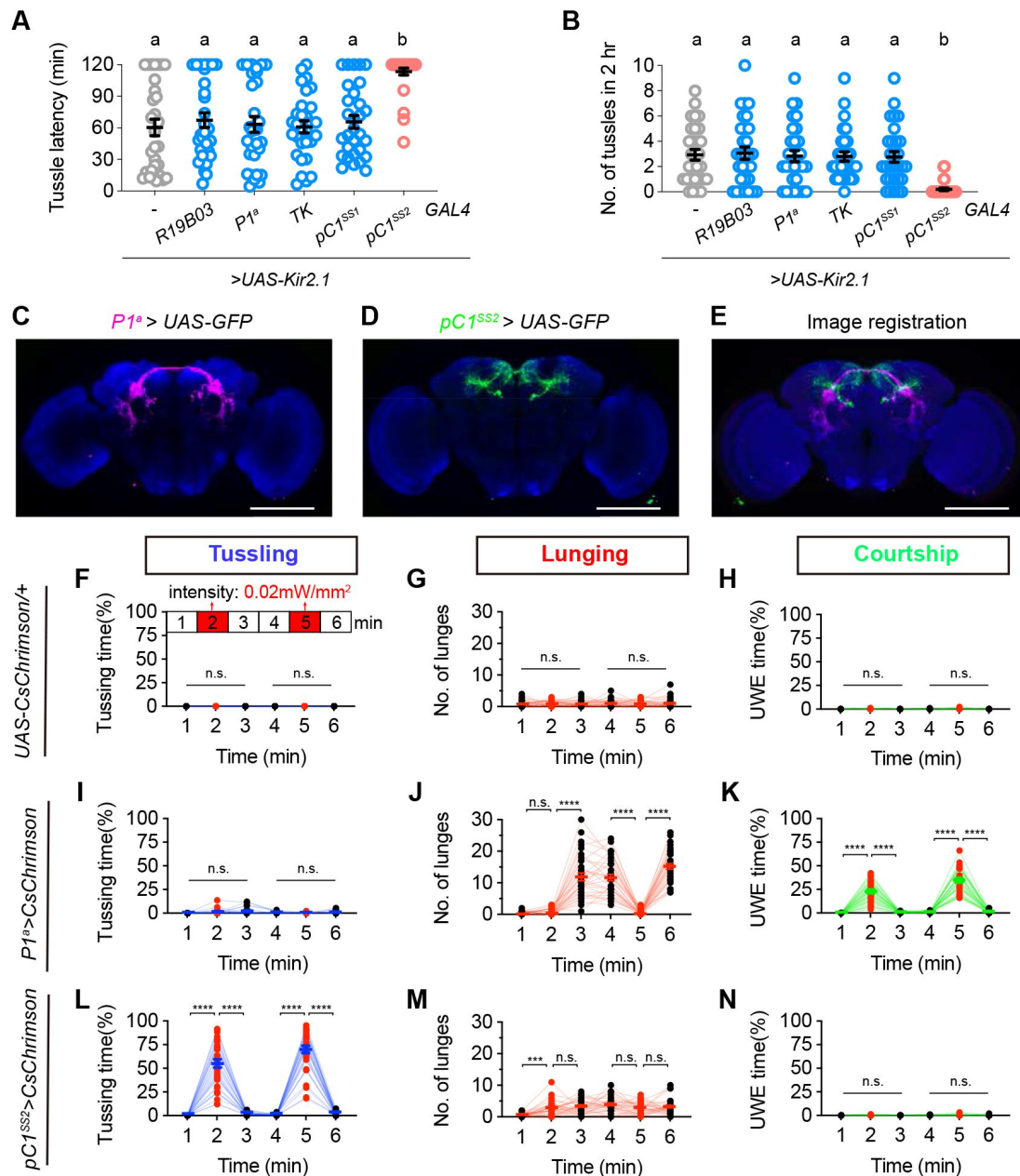


Figure 4.

A subset of pC1 neurons specifically promote male tussling behavior.

(A, B) The effect of inactivating candidate interneurons on latency (A) and number (B) of male tussling behavior, $n=30$ for each group, bars sharing the same letter are not significantly different according to Kruskal-Wallis test.

(C-E) The expression pattern of *P1^a-spGAL4* (C) and *pC1^{SS2}-spGAL4* (D) in male brain, and (E) the registration of *P1^a* (magenta) and *pC1^{SS2}* (green) neurons in a standard brain. The GFP expression indicates *P1^a* (magenta) and *pC1^{SS2}* (green) neurons in the male brain counterstained with *nc82* (blue). Scale bars, 100 μ m.

(F-H) Statistics on tussling (F), lunging (G) and unilateral wing extension (UWE, H) by *UAS-CsChrimson/+* males. Red light activation with a light intensity of 0.02 mW/mm² for 1 minute was applied at the 2nd and 5th minutes of this and subsequent experiments. $n=32$ for each group, n.s., not significant, Friedman test.

(I-K) Statistics on tussling (I), lunging (J) and UWE (K) in *P1^a>UAS-CsChrimson* males. $n=32$ for each group, n.s., not significant, ****P < 0.0001, Friedman test.

(L-N) Statistics on tussling (L), lunging (M) and UWE (N) in *pC1^{SS2}>UAS-CsChrimson* males. $n=32$ for each group, n.s., not significant, ****P < 0.0001, ****P < 0.0001, Friedman test.

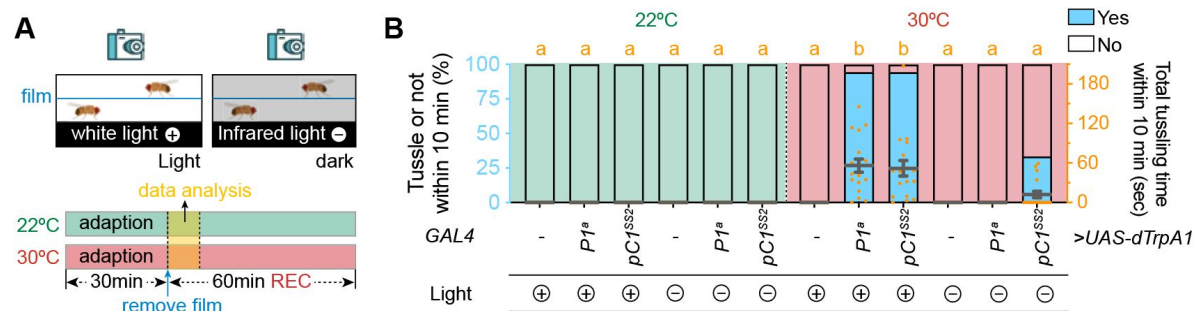


Figure 4—figure supplement 1.

Light-dependent initiation of male tussling behavior.

(A) Schematic diagram of the thermal activation experiment.

(B) The percentage of tussling occurrence (left) and total tussling time (right) within 10 mins. $n=18$ for each group, bars sharing the same letter are not significantly different according to Kruskal-Wallis test based on the data of total tussling time.

Previous studies have revealed a crucial role of $P1^a$ neurons in promoting a persistent internal state that enhances male aggression. To test whether $P1^a$ neurons can promote both lunging and tussling behaviors, we further performed thermal activation experiments. We found that activation of $P1^a$ neurons via dTrpA1 at 30°C induced male tussling, just like activation of $pC1^{SS2}$ neurons did (**Figure 4—figure supplement 1A** [↗](#) and **1B** [↗](#)). As thermal activation but not optogenetic activation of $P1^a$ neurons induced tussling, we reasoned that vision might be a key factor for this discrepancy, since all optogenetic experiments were performed in dark or red light. Thus, we re-performed these thermal activation experiments in dark and found that activation of $pC1^{SS2}$ but not $P1^a$ neurons induced tussling, although the level of tussling by $pC1^{SS2}$ -activated males was lower compared to that in light, suggesting a crucial role of vision in male tussling. These optogenetic and thermogenetic activation experiments suggest that $P1^a$ neurons can promote both lunging and tussling behaviors, but their function on tussling requires visual stimulation, while $pC1^{SS2}$ neurons play a more direct role in acutely promoting male tussling.

Territory and Mating Advantage in Tussling-favored Experienced Males

As our results showed that GH males and SH males were more likely to perform the low-frequency, high-intensity tussling and the high-frequency, low intensity lunging, respectively, we wondered which fighting strategy was more practical in the competition for territory and mating resources.

To address this question, we first designed an experimental paradigm for territory competition (**Figure 5A** [↗](#)). A circular territory, which is provided with food and a fixed virgin female and separated by a wall with a small door open to the outer area, was competed by two tester males. We manually analyzed the first 10 winning events; each was defined as one male successfully displacing the other from the inner resource area. The winning index was calculated as the proportional difference in the number of wins between each pair of males. In control groups such as two GH males of the same age, we observed some ‘winner takes all’ phenomenon (one male won nearly all of the 10 encounters), but overall, the winning index was close to zero. We further found that while GH 7-day-old males (G7) won slightly, but not significantly, more than SH 7-day-old males (S7) (**Figure 5B** [↗](#)), older G14 males had a significant advantage over the S14 males (**Figure 5C** [↗](#) and **Figure 5—figure supplement 1** [↗](#), see Video 3).

Next, we conducted further investigations into the role of social experience and the two types of fighting behaviors in mating competition (**Figure 5D** [↗](#)). Two males and one virgin female were loaded into the behavioral chamber at the same time and recorded for courtship behaviors. We found that G7 males had no significant advantage over S7 males (**Figure 5E** [↗](#)); however, G14 males displayed a significantly increased mating advantage over S14 males (**Figure 5F** [↗](#), see Video 4). Together, these results suggest that the tussling-favored GH males are more competitive in territory control and mating compared to the lunging-favored SH males.

The above results showed that GH males were more competitive compare to SH males, especially in older males, suggesting that social experience may play a potential role in altering age-related mating competition. To test this hypothesis, we first tested mating competition between males of different ages under the same rearing conditions. The results showed that younger males, whether reared in group or isolation, had a greater mating advantage compared to older males (G7 > G14, S7 > S14, G14 > G21, S14 > S21) (**Figure 5G** [↗](#)). These results indicate that there is generally a disadvantage in mating competition associated with aging. We next investigated whether the mating advantage induced by social experience could compensate for the mating disadvantage caused by aging. We tested pairs of males with different ages and rearing conditions (G7 vs. S14, S7 vs. G14, G14 vs. S21, S14 vs. G21) for mating competition. We found that while 14-day-old males were less competitive than 7-day-old males regardless of the rearing condition, indicative of the aging-related disadvantage, 21-day-old GH males were significantly more competitive than 14-day-

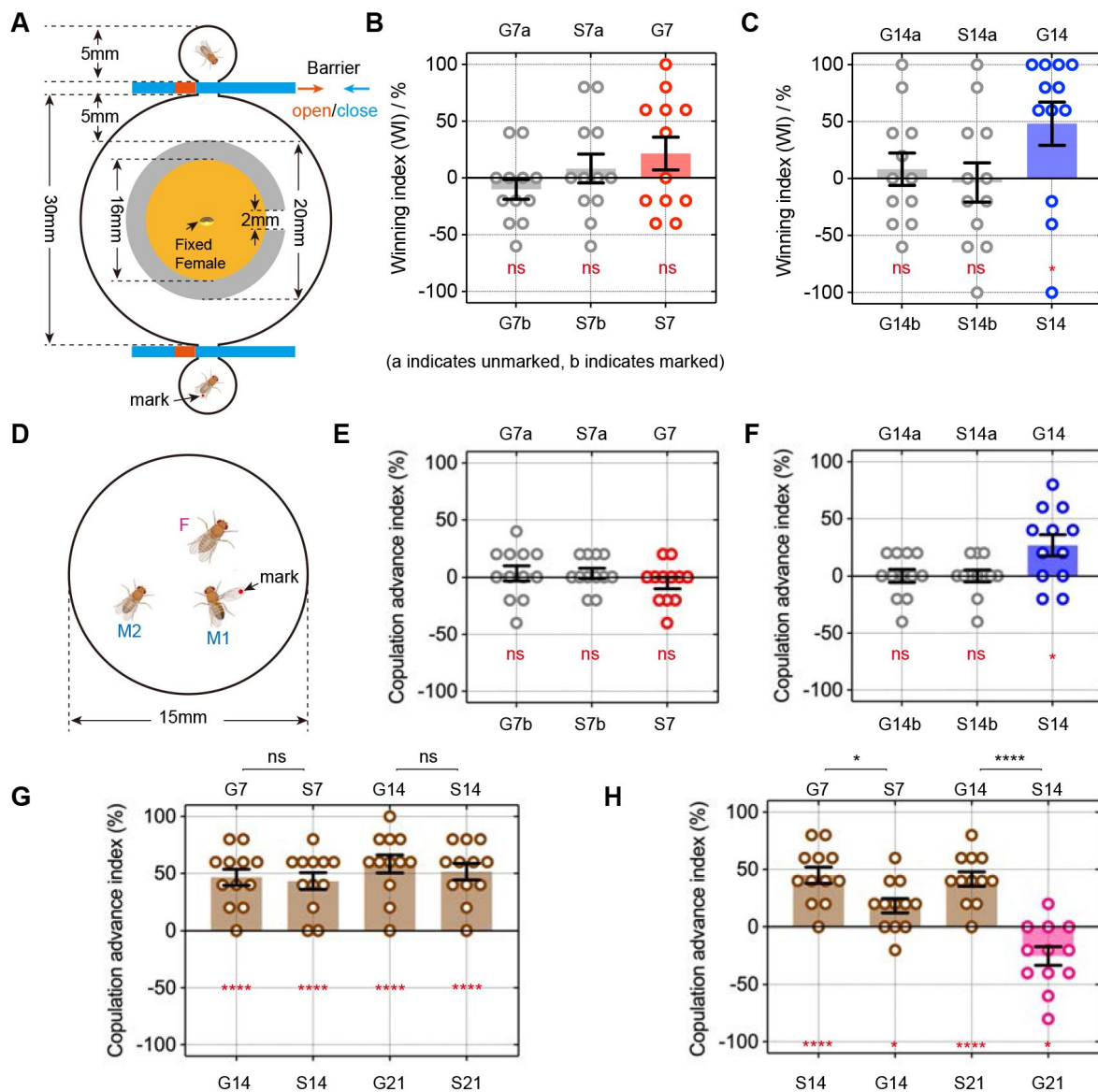


Figure 5.

Social enrichment enhances male territorial control and mating competition.

(A) Schematic diagram of the paradigm for testing territorial control in males.

(B) The territorial control test of 7-day-old GH (G7) and SH (S7) males. G7a and G7b stand for marked G7 male and unmarked G7 male, respectively. $n=12$ for each group, n.s., not significant, one sample t test.

(C) The territorial control test of 14-day-old GH (G14) and SH (S14) males. $n=12$ for each group, n.s., not significant, * $P < 0.05$, one sample t test.

(D) Schematic diagram of the paradigm for testing male mating advantage.

(E) The mating competition test between G7 and S7 males. $n=12$ for each group, n.s., not significant, one sample t test.

(F) The mating competition test between G14 and S14 males. $n=12$ for each group, n.s., not significant, * $P < 0.05$, one sample t test.

(G and H) The mating competition test between young males and old males, $n=12$ for each group, n.s., not significant, * $P < 0.05$, **** $P < 0.0001$, one sample t test for comparison within group (red labeling), Mann-Whitney test for comparison between groups (black labeling). Error bars correspond to SEM.

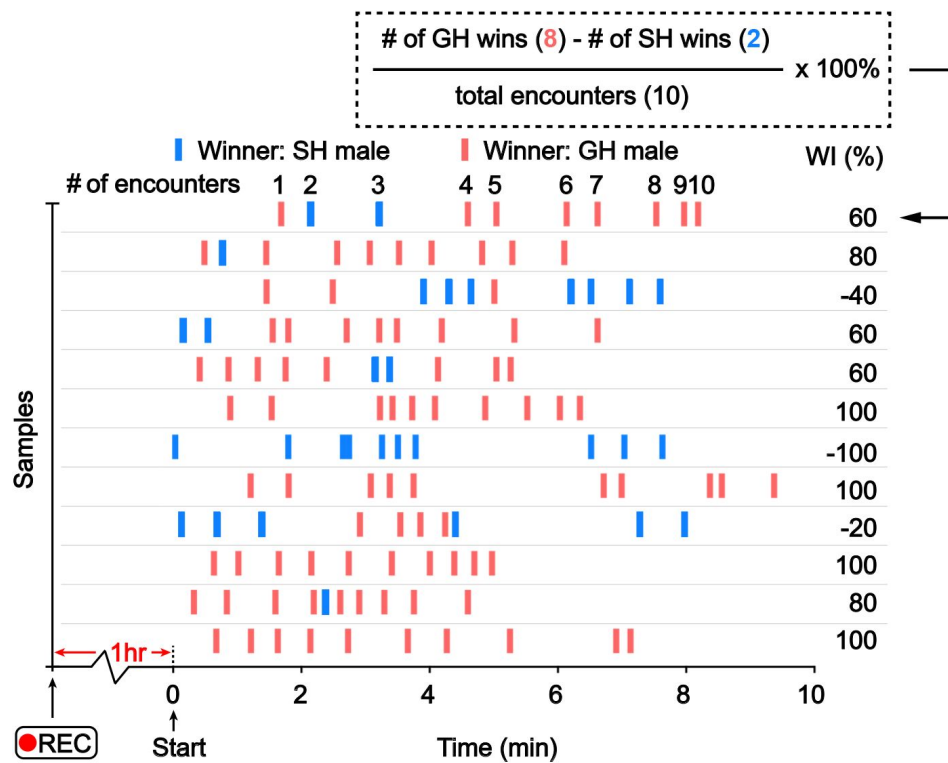


Figure 5—figure supplement 1.

Winning events of representative samples for territorial competition.

The blue and red bars represent the wins for S14 males and G14 males, respectively. An example formula is provided for calculating the winning index in the dashed box.

old SH males (G21 > S14) (**Figure 5H** [↗](#)). These results indicate that long-term social enrichment in older males increases their territorial and mating competition, thus overcoming their mating disadvantage associated with aging (**Figure 6** [↗](#)).

Discussion

Regulation of physiology and behavior by social experience is an important feature across animal species. Here, we found that social experience shapes the fighting strategies of male flies, in which GH males tend to reduce the low-intensity lunging and increase the high-intensity tussling, which enhances their territorial and mating competition. We further showed that the two forms of fighting are mediated by distinct sensory and central neurons. The complex implications of these findings are discussed below.

Previous studies have found that social isolation generally enhances aggression but decreases mating competition in animal models, presenting a paradox between social experience, aggression and reproductive success. Our results resolved this paradox by dissecting aggressive behaviors into the well-studied, frequently observed lunging behavior and the less frequent but high-intensity tussling behavior. We established a new behavioral paradigm for studying tussling behavior and revealed that social experience oppositely regulates lunging and tussling. It is important to note that the frequency of lunging decreases while the frequency of tussling increases from 7-day-old to 14-day-old males, suggesting a shift in fighting strategy with age. Our results further suggest that social experience may enhance male mating competition by shifting their fighting strategy to the low-frequency, high-intensity tussling, which may be more effective in combat, although direct evidence is still lacking. Interestingly, we found that the mating advantage gained through social enrichment can even offset the mating disadvantage associated with aging, further supporting the vital role of shifting fighting strategies in experienced, aged males. This dissection of the two fighting forms rectifies our understanding of how social experiences regulates aggression and reproductive success.

Our results revealed distinct neural circuits underlying the two forms of fighting, spanning from primary sensory neurons to central regulatory neurons. Specifically, silencing the cVA-sensing Or67d neurons reduced male lunging but not tussling, whereas inactivating Or47b neurons impaired male tussling but not lunging. Notably, Or47b neurons respond to pheromones common to both sexes, and the sensitivity of these sensory neurons is enhanced by social enrichment [[18](#) [↗](#), [19](#) [↗](#), [21](#) [↗](#)]. Unlike the Or67d-dependent lunging, which is primarily male-induced and socially inhibited, the Or47b-dependent tussling is promoted by social enrichment with either male or female flies. Recently, it has been found that female flies increase aggression towards mating pairs or mated females, depending on Or47b and Or67d, respectively [[32](#) [↗](#), [33](#) [↗](#)]. These results suggest a common function of Or47b and Or67d neurons in both sexes for regulating aggression. However, whether these neurons mediate different fighting patterns in females and how they respond to social experience in females remain unknown. Nevertheless, our results on these two sensory pathways in males provide crucial evidence for dissecting their functions.

Regarding central regulatory neurons, we identified a novel class of pC1 neurons, pC1^{SS2}, that specifically promote male tussling. In male flies, there are ~60 pairs of *dsx*-expressing pC1 neurons, including the well-characterized P1^a neurons that also express *fru*^M [[34](#) [↗](#), [35](#) [↗](#)]; however, there are only three pairs of pC1^{SS2} neurons whose function is both necessary and sufficient for inducing tussling in males. The three pairs of pC1^{SS2} neurons may have distinct function from P1^a neurons in several ways. (1) P1^a neurons promotes both male courtship and aggression [[36](#) [↗](#), [37](#) [↗](#)], whereas pC1^{SS2} neurons specifically promote aggression but not courtship. A recent study also identified pC1^{SS2} neurons as aggression-promoting, but their roles in different fighting modes are not depicted [[38](#) [↗](#)]. (2) P1^a neurons promotes both lunging and tussling, although their function on tussling requires additional visual stimulation. In contrast,

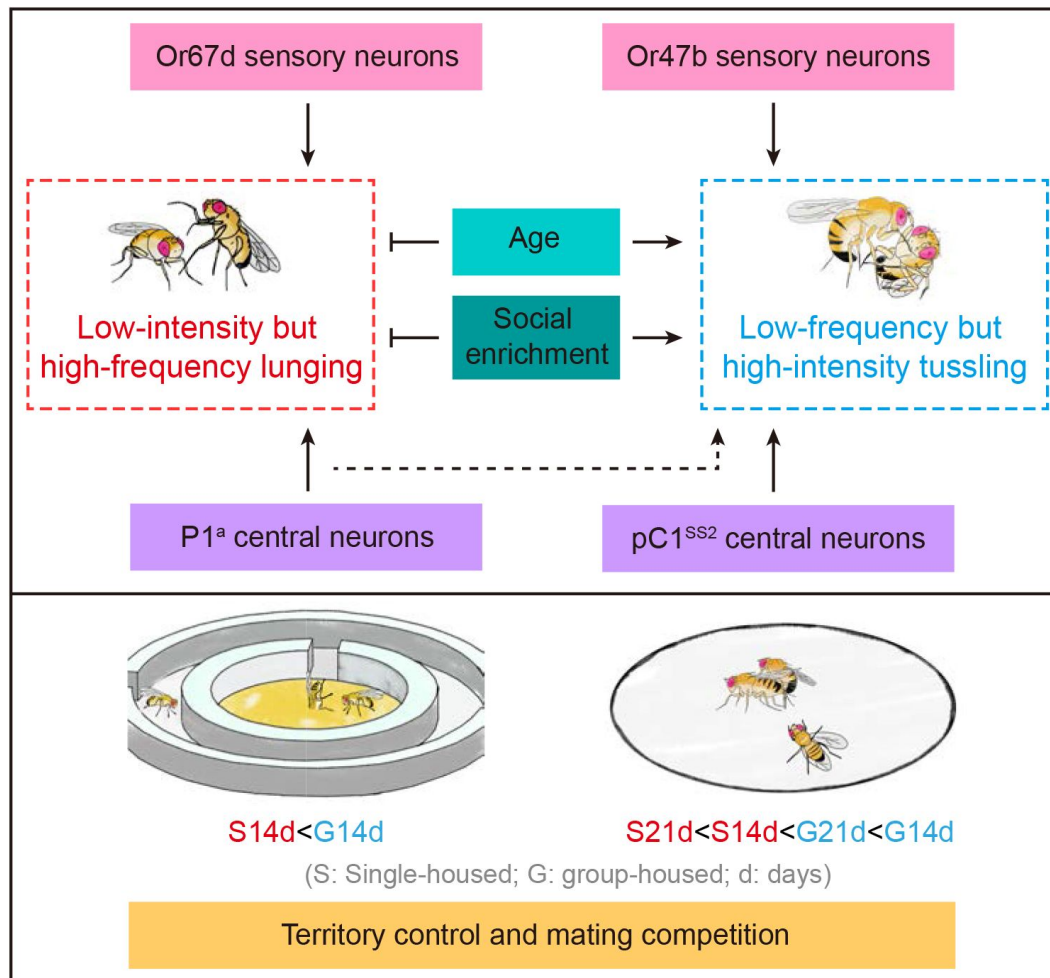


Figure 6.

Summary model of fighting strategies and reproductive success in male *Drosophila*.

Age and social enrichment inhibit the low-intensity, high-frequency lunging while promoting the low-frequency, high-intensity tussling. This shift in fighting strategies among experienced, aged males enhances their territorial control and mating competition, even offsetting aging-related mating disadvantages. Lunging and tussling behaviors are regulated by distinct sensory and central neurons.

pC1^{SS2} neurons specifically promote tussling, and the presence of light enhances but is not absolutely necessary for tussling. These results align with previous findings that visual stimuli amplify P1 activation and related behaviors [39–41], and visual stimulation is important for male aggression [42, 43]. (3) Male lunging is induced after P1^a activation; whereas male tussling is induced during pC1^{SS2} activation. These results suggest that P1^a neurons may not directly regulate aggression but instead induce a persistent state that promote aggression in general, while pC1^{SS2} neurons may play a more direct role in mediating tussling. Future studies are needed to further dissect the seemingly parallel sensory and central neuronal pathways involved in lunging and tussling, and to explore how these pathways interact during social experience and aging to optimize reproductive success.

Materials and methods

Fly Stocks

Flies were maintained at 22 or 25 °C in a 12 h:12 h light: dark cycle. *Canton-S* flies were used as the wild-type strain. *Or67d-GAL4* (BDSC_9998), *Or65a-GAL4* (BDSC_9993), *Or47b-GAL4* (BDSC_9984), *IR84a-GAL4* (BDSC_41734), *TK-GAL4* (BDSC_51795), *R19B03-GAL4* (BDSC_49830), *UAS-dTrpA1* (BDSC_26263), *UAS-Kir2.1* (BDSC_6595), *UAS-CsChrimson* (BDSC_82181), *UAS-myrGFP* (BDSC_32198) and *trans-Tango* (BDSC_77124) flies were obtained from Bloomington *Drosophila* Stock Center (BDSC). *UAS-NaChBac* [30], *fru^{GAL4}* [44], *P1^a-spGAL4* [45], *pC1^{SS1}-spGAL4* [46], *pC1^{SS2}-spGAL4* [46] and *UAS-fruMi* [47] were used as previously described. *UAS-Or47b RNAi* (THU_2599) was obtained from TsingHua Fly Center.

Male aggression assay

For the aggression assays presented in **Figure 1**, the prepared food was evenly spread into the chambers of two paradigms. There was no female in the traditional paradigm, while a 3-5 day old virgin female was gently fixed in the center of the food with its abdomen outside in the new paradigm. Then two male testers were placed into the upper chambers separated from the food (with or without a fixed female) by a film barrier. All testers were loaded by cold anesthesia. After a 30-minute adaptation, the film was removed, and the aggressive behavior was recorded for 2 hours. The aggression assays in **Figure 2** and **3** were performed by using the new paradigm. The aggression assays in **Figure 1—figure supplement 1** was performed by using the traditional paradigm.

The food recipes for the male aggression assay were listed below. A mixture of 2.5% yeast, 2.5% sucrose and 1% agar dissolved in apple juice was used for **Figure 1B**; a mixture of 2.5% yeast, different concentrations of sucrose (0, 2.5%, 5% or 10%) and 1% agar dissolved in apple juice was used for **Figure 1C**; a mixture of different concentrations of yeast (0, 2.5%, 5% or 10%), 2.5% sucrose and 1% agar dissolved in apple juice was used for **Figure 1D**; A mixture of 10% yeast and 1% agar dissolved in apple juice was used for other experiments.

dTrpA1-mediated activation experiments

For thermogenetic activation assay showed in **Figure 4—figure supplement 1**, two 5-7 days old males were loaded into the circular chamber (d=15mm) under cold anesthesia and separated with a transparent film. Before the recording start, the chamber was transferred to the recording platform at a specific temperature (22°C or 30°C) for 30 minutes adaptation. After that, the film was removed, and the behavior was recorded for 1 hour. The 10-minute video after film extraction was used for behavioral analysis.

CsChrimson-mediated optogenetic activation experiments

For optogenetic activation assay showed in **Figure 4**, males were raised at 25°C on standard fly food, collected after eclosion and housed in the dark for 5-7 days on food containing 0.4 mM all *trans*-Retinal (MFCD00001550, Sigma-Aldrich, St. Louis, MO). The specific stimulation protocols are showed in the schematic (**Figure 4F**). In brief, Two 5-7 days old males were loaded into the circular chamber (d=10mm) under freeze anesthesia and separated with a transparent film in the dark. Then the chamber was moved on the plane light source (IR channel: 865nm and red-light channel: 630nm) for 15 mins adaptation. Before video recording, the IR light channel was switched on and adjusted to the appropriate brightness for recording. The film was removed before the recording start, with the red-light channel switched on (constant red light, 0.02 mW/mm²) during the 2nd and 5th minutes of the experiment. The statistics for the tussling, lunging, and unilateral wing extension (UWE) are as follows. For tussling time and UWE time, we used LifesongX software to calculate the percentage of the total time these behaviors occurred per minute. For lunges, we counted the number of lunging occurrences per minute.

Territorial competition experiment

For the territorial competition experiment in **Figures 5A-C**, the behavioral setup was produced using 3D printing technology, as illustrated in the schematic, with a transparent cover plate placed on the top. The preparation of food and female were the same as the new aggression paradigm. The female was fixed in the center of the food and two male testers were placed separately in the two small chambers on both sides. Male wings were marked 24 hours before the experiment. Before initiating the experiment, the two barriers were closed. After a 30-minute adaptation, the barriers were opened to allow the males access to the central large chamber, and were closed once the males went inside, and the territorial behavior was recorded for 2 hours. The winning index stand for territorial control ability was calculated by the following example formula: (num. of wins by GH males – num. of wins by SH males)/10 encounters * 100%. The first 10 encounters occurring after 1 hour of recording were used for statistical analysis. The winning event was defined as a male driving the other male out of the food area.

Mating competition experiment

For the mating competition experiment shown in **Figure 5D-H**, the wings of males were marked 24 hours prior to the experiment using the same method described above. During the experiment, a 3-5 days old virgin female was placed into the chamber under freeze anesthesia and covered with a transparent film. Then the unmarked males were loaded into the chamber in the same manner and also covered with a transparent film. Finally, the marked males were loaded. After 30 minutes adaptation, two films were removed, and the behavior was recorded for 1 hour. The chambers without successful copulation during the 1-hour period were excluded from further analysis. Each assay included 120 efficient chambers at least, and 10 random chambers were measured as one sample. The copulation advance index was calculated by the following example formula: (num. of GH winners – num. of SH winners)/10 chambers * 100%). The winner was defined as the male that completed the first copulation with female.

Quantitative Real-time PCR

Adult fly samples were frozen in liquid nitrogen and decapitated by vortex. The heads were then separated from the bodies using metal sieves. Each sample, consisting of 40 frozen heads, was used for total RNA preparation using TRIzol reagent (15596026, Thermo Fisher Scientific, Waltham, MA). We purified total RNA using a DNA-freeTM Kit (AM1906, Thermo Fisher Scientific, Waltham, MA) and performed reverse transcription using SuperScriptTM IV (18091050, Thermo Fisher Scientific, Waltham, MA) to obtain cDNA used for templates. Quantitative PCR was performed on the Roche LightCycler[®] 96 Real-Time PCR machine using AceQ qPCR SYBR Green

Master Mix (Q121-02, Vazyme, Nanjing). Transcript levels were analyzed by the $2^{-\Delta\text{CT}}$ method using *actin* as control. Each sample was run in triplicate. Each experiment was repeated three times using independent sets of genetic crosses. Primers used for RT-qPCR quantification were:

actin forward: 5'-GTCGCGATTTAACCGACTACCTGA-3'

actin reverse: 5'-TCTTGCTT CGAGATCCACATCTGC-3'

fru-P1 forward: 5'-GTGTGCGTACGTTTGAGTGT-3'

fru-P1 reverse: 5'-TAATCCTGTGACGTCGCCAT-3'

Or47b forward: 5'-CAAATCTCAGCCTTCTGCGG-3'

Or47b reverse: 5'-GATACTGGCACAGCAAACCTCA-3'

Tissue Dissection, Staining, and Imaging

Brains were dissected in Schneider's insect medium (S2) and fixed in 4% paraformaldehyde in 0.5% Triton X-100 and 0.5% bovine serum albumin in phosphate-buffered saline (PAT) for 30 min at room temperature. After 4×10 -min washes, tissues were blocked in 3% normal goat serum (NGS) for 60 min, then incubated in primary antibodies diluted in 3% NGS for 4 h at room temperature and 1 day at 4°C, then washed in PAT, and incubated in secondary antibodies diluted in 3% NGS for 4 h at room temperature and 1 day at 4°C. The tissues were then washed thoroughly in PAT and mounted for imaging. The antibodies used were Rabbit anti-HA (1:1000), Mouse anti-GFP (1:1000), nc82(1:50) and Donkey anti-Mouse, Alexa Fluor 488 (1:500), Donkey anti-Rabbit, Alexa Fluor 555 (1:500). Samples were imaged on Zeiss 900 confocal microscopes using ZEN and processed with Fiji software.

Brain image registration

The protocol for standard brain registration is described previously [48]. Confocal images for brains containing P1^a neurons and pC1^{SS2} neurons were registered onto a standard reference brain, which was generated by the images of nc82 stained brains.

Statistics

Experimental flies and genetic controls were tested under the same conditions, and data were collected from at least two independent experiments. Statistical analysis was performed using GraphPad Prism 9 (<https://www.graphpad.com/scientific-software/prism/>) as indicated in each figure legend. Data were first verified for normal distribution by the D'Agostino–Pearson normality test. For normally-distributed data, Student's *t* test was used for pairwise comparisons, and one-way ANOVA was used for comparisons among multiple groups, followed by Tukey's multiple comparisons. For data not normally-distributed, the Mann-Whitney test was used for pairwise comparisons, and the Kruskal–Wallis test was used for comparison among multiple groups, followed by Dunn's multiple comparisons. For group matched data that were normally-distributed, the Paired *t* test signed rank test was used for pairwise comparisons, and RM one-way ANOVA was used for comparison among multiple groups, followed by RM one-way ANOVA multiple comparisons. For group matched data that were not normally-distributed, the Wilcoxon matched-pairs signed rank test was used for pairwise comparisons, and the Friedman test was used for comparison among multiple groups, followed by Dunn's multiple comparisons. For qPCR experiments, the average relative expression of three independent experiments was analyzed using Mann-Whitney test. For one group compared with zero, we used one sample *t* test. The Chi-square test was used to compare male tussling percentage between two groups.

Data availability

All data generated or analyzed during this study are included in the article and its supporting information files. Fly stocks and reagents used in this study are available from the corresponding author upon reasonable request.

Additional information

Author contributions

C. Gao, Q. Peng and Y. Pan conceived the study; C. Gao and M. Ma performed all experiments and analyzed the data with help from J. Chen and X. Ji; Q. Peng and Y. Pan supervised the study; C. Gao and Y. Pan wrote the manuscript with input from all authors.

Competing interests

The authors declare no competing interests.

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Reviewer #1 (Public review):

This work addresses an important question in the field of *Drosophila* aggression and mating-prior social isolation is known to increase aggression in males by increased lunging, which is suppressed by group housing (GH). However, it is also known that single-housed (SH) males, despite their higher attempts to court females, are less successful. Here, Gao et al., developed a modified aggression assay, to address this issue by recording aggression in *Drosophila* males for 2 hours, over a virgin female which is immobilized by burying its head in the food. They found that while SH males frequently lunge in this assay, GH males switch to higher intensity but very low-frequency tussling. Constitutive neuronal silencing and activation experiments implicate cVA sensing Or67d neurons promoting high-frequency lunging, similar to earlier studies, whereas Or47b neurons promote low-frequency but higher intensity tussling. Using optogenetic activation they found that three pairs of pC1 neurons- pC1SS2 increase tussling. While P1a neurons, previously implicated in promoting aggression and courtship, did not increase tussling in optogenetic activation (in the dark), they could promote aggressive tussling in thermogenetic activation carried out in the presence of visible light. It was further suggested, using a further modified aggression assay that GH males use increased tussling and are able to maintain territorial control, providing them mating advantage over SI males and this may partially overcome the effect of aging in GH males.

Strengths:

Using a series of clever neurogenetic and behavioral approaches, subsets of ORNs and pC1 neurons were implicated in promoting tussling behaviors. The authors devised a new paradigm to assay for territory control which appears better than earlier paradigms that used a food cup (Chen et al, 2002), as this new assay is relatively clutter-free, and can be eventually automated using computer vision approaches. The manuscript is generally well-written, and the claims made are largely supported by the data.

Weaknesses:

I have a few concerns regarding some of the evidence presented and claims made as well as a description of the methodology, which needs to be clarified and extended further.

(1) Typical paradigms for assaying aggression in *Drosophila* males last for 20-30 minutes in the presence of nutritious food/yeast paste/females or all of these (Chen et al. 2002, Nilsen et al., 2004, Dierick et al. 2007, Dankert et al., 2009, Certel & Kravitz 2012). The paradigm described in Figure 1 A, while important and more amenable for video recording and computational analysis, seems a modification of the assay from Kravitz lab (Chen et al., 2002), which involved using a female over which males fight on a food cup. The modifications include a flat surface with a central food patch and a female with its head buried in the food, (fixed female) and much longer adaptation and recording times respectively (30 minutes, 2 hours), so in that sense, this is not a 'new' paradigm but a modification of an existing paradigm and its description as new should be appropriately toned down. It would also be important to cite these earlier studies appropriately while describing the assay.

(2) Lunging is described as a 'low intensity' aggression (line 111 and associated text), however, it is considered a mid to high-intensity aggressive behavior, as compared to other lower-intensity behaviors such as wing flicks, chase, and fencing. Lunging therefore is lower in intensity 'relative' to higher intensity tussling but not in absolute terms and it should be mentioned clearly.

- (3) It is often difficult to distinguish faithfully between boxing and tussling and therefore, these behaviors are often clubbed together as box, tussle by Nielsen et al., 2004 in their Markov chain analysis as well as a more detailed recent study of male aggression (Simon & Heberlein, 2020). Therefore, authors can either reconsider the description of behavior as 'box, tussle' or consider providing a video representation/computational classifier to distinguish between box and tussle behaviors.
- (4) Simon & Heberlein, 2020 showed that increased boxing & tussling precede the formation of a dominance hierarchy in males, and lunges are used subsequently to maintain this dominant status. This study should be cited and discussed appropriately while introducing the paradigm.
- (5) It would be helpful to provide more methodological details about the assay, for instance, a video can be helpful showing how the males are introduced in the assay chamber, are they simply dropped to the floor when the film is removed after 30 minutes (Figures 1-2)?
- (6) The strain of Canton-S (CS) flies used should be mentioned as different strains of CS can have varying levels of aggression, for instance, CS from Martin Heisenberg lab shows very high levels of aggressive lunges. Are the CS lines used in this study isogenized? Are various genetic lines outcrossed into this CS background? In the methods, it is not clear how the white gene levels were controlled for various aggression experiments as it is known to affect aggression (Hoyer et al. 2008).
- (7) How important it is to use a fixed female for the assay to induce tussling? Do these females remain active throughout the assay period of 2.5 hours? Is it possible to use decapitated virgin females for the assay? How will that affect male behaviors?
- (8) Raster plots in Figure 2 suggest a complete lack of tussling in SH males in the first 60 minutes of the encounter, which is surprising given the longer duration of the assay as compared to earlier studies (Nielsen et al. 2004, Simon & Heberlein, 2020 and others), which are able to pick up tussling in a shorter duration of recording time. Also, the duration for tussling is much longer in this study as compared to shorter tussles shown by earlier studies. Is this due to differences in the paradigm used, strain of flies, or some other factor? While the bar plots in Figure 2D show some tussling in SH males, maybe an analysis of raster plots of various videos can be provided in the main text and included as a supplementary figure to address this.
- (9) Neuronal activation experiments suggesting the involvement of pC1SS2 neurons are quite interesting. Further, the role of P1a neurons was demonstrated to be involved in increasing tussling in thermogenetic activation in the presence of light (Figure 4, Supplement 1), which is quite important as the role of vision in optogenetic activation experiments, which required to be carried out in dark, is often not mentioned. However, in the discussion (lines 309-310) it is mentioned that PC1SS2 neurons are 'necessary and sufficient' for inducing tussling. Given that P1a neurons were shown to be involved in promoting tussling, this statement should be toned down.
- (10) Are Or47b neurons connected to pC1SS2 or P1a neurons?
- (11) The paradigm for territory control is quite interesting and subsequent mating advantage experiments are an important addition to the eventual outcome of the aggressive strategy deployed by the males as per their prior housing conditions. It would be important to comment on the 'fitness outcome' of these encounters. For instance, is there any fitness advantage of using tussling by GH males as compared to lunging by SH males? The authors may consider analyzing the number of eggs laid and eclosed progenies from these encounters to address this.

Reviewer #2 (Public review):

Summary:

Gao et al. investigated the change of aggression strategies by the social experience and its biological significance by using *Drosophila*. Two modes of inter-male aggression in *Drosophila* are known: lunging, high-frequency but weak mode, and tussling, low-frequency but more vigorous mode. Previous studies have mainly focused on the lunging. In this paper, the authors developed a new behavioral experiment system for observing tussling behavior and found that tussling is enhanced by group rearing while lunging is suppressed. They then searched for neurons involved in the generation of tussling. Although olfactory receptors named Or67d and Or65a have previously been reported to function in the control of lunging, the authors found that these neurons do not function in the execution of tussling, and another olfactory receptor, Or47b, is required for tussling, as shown by the inhibition of neuronal activity and the gene knockdown experiments. Further optogenetic experiments identified a small number of central neurons pC1[SS2] that induce the tussling specifically. In order to further explore the ecological significance of the aggression mode change in group rearing, a new behavioral experiment was performed to examine territorial control and mating competition. Finally, the authors found that differences in the social experience (group vs. solitary rearing) are important in these biologically significant competitions. These results add a new perspective to the study of aggressive behavior in *Drosophila*. Furthermore, this study proposes an interesting general model in which the social experience-modified behavioral changes play a role in reproductive success.

Strengths:

A behavioral experiment system that allows stable observation of tussling, which could not be easily analyzed due to its low frequency, would be very useful. The experimental setup itself is relatively simple, just the addition of a female to the platform, so it should be applicable to future research. The finding about the relationship between the social experience and the aggression mode change is quite novel. Although the intensity of aggression changes with the social experience was already reported in several papers (Liu et al., 2011, etc), the fact that the behavioral mode itself changes significantly has rarely been addressed and is extremely interesting. The identification of sensory and central neurons required for the tussling makes appropriate use of the genetic tools and the results are clear. A major strength of the neurobiology in this study is the finding that another group of neurons (Or47b-expressing olfactory neurons and pC1[SS2] neurons), distinct from the group of neurons previously thought to be involved in low-intensity aggression (i.e. lunging), function in the tussling behavior. Further investigation of the detailed circuit analysis is expected to elucidate the neural substrate of the conflict between the two aggression modes.

Weaknesses:

The experimental systems examining the territory control and the reproductive competition in Figure 5 are novel and have advantages in exploring their biological significance. However, at this stage, the authors' claim is weak since they only show the effects of age and social experience on territorial and mating behaviors, but do not experimentally demonstrate the influence of aggression mode change itself. In the Abstract, the authors state that these findings reveal how social experience shapes fighting strategies to optimize reproductive success. This is the most important perspective of the present study, and it would be necessary to show directly that the change of aggression mode by social experience contributes to reproductive success.

In addition, a detailed description of the tussling is lacking. For example, the authors state that the tussling is less frequent but more vigorous than lunging, but while experimental data are presented on the frequency, the intensity seems to be subjective. The intensity is certainly clear from the supplementary video, but it would be necessary to evaluate the intensity itself using some index. Another problem is that there is no clear explanation of how to determine the tussling. A detailed method is required for the reproducibility of the experiment.

<https://doi.org/10.7554/eLife.104212.1.sa2>

Reviewer #3 (Public review):

In this manuscript, Gao et al. presented a series of intriguing data that collectively suggest that tussling, a form of high-intensity fighting among male fruit flies (*Drosophila melanogaster*) has a unique function and is controlled by a dedicated neural circuit. Based on the results of behavioral assays, they argue that increased tussling among socially experienced males promotes access to resources. They also concluded that tussling is controlled by a class of olfactory sensory neurons and sexually dimorphic central neurons that are distinct from pathways known to control lunges, a common male-type attack behavior.

A major strength of this work is that it is the first attempt to characterize the behavioral function and neural circuit associated with *Drosophila* tussling. Many animal species use both low-intensity and high-intensity tactics to resolve conflicts. High-intensity tactics are mostly reserved for escalated fights, which are relatively rare. Because of this, tussling in the flies, like high-intensity fights in other animal species, has not been systematically investigated. Previous studies on fly aggressive behavior have often used socially isolated, relatively young flies within a short observation duration. Their discovery that 1) older (14-days-old) flies tend to tussle more often than younger (2-days-old) flies, 2) group-reared flies tend to tussle more often than socially isolated flies, and 3) flies tend to tussle at a later stage (mostly ~15 minutes after the onset of fighting), are the result of their creativity to look outside of conventional experimental settings. These new findings are keys for quantitatively characterizing this interesting yet under-studied behavior.

Precisely because their initial approach was creative, it is regrettable that the authors missed the opportunity to effectively integrate preceding studies in their rationale or conclusions, which sometimes led to premature claims. Also, while each experiment contains an intriguing finding, these are poorly related to each other. This obscures the central conclusion of this work. The perceived weaknesses are discussed in detail below.

Most importantly, the authors' definition of "tussling" is unclear because they did not explain how they quantified lunges and tussling, even though the central focus of the manuscript is behavior. Supplemental movies S1 and S2 appear to include "tussling" bouts in which 2 flies lunge at each other in rapid succession, and supplemental movie S3 appears to include bouts of "holding", in which one fly holds the opponent's wings and shakes vigorously. These cases raise a concern that their behavior classification is arbitrary. Specifically, lunges and tussling should be objectively distinguished because one of their conclusions is that these two actions are controlled by separate neural circuits. It is impossible to evaluate the credibility of their behavioral data without clearly describing a criterion of each behavior.

It is also confusing that the authors completely skipped the characterization of the tussling-controlling neurons they claimed to have identified. These neurons (a subset of so-called pC1 neurons labeled by previously described split-GAL4 line pC1SS2) are central to this manuscript, but the only information the authors have provided is its gross morphology in a low-resolution image (Figure 4D, E) and a statement that "only 3 pairs of pC1SS2 neurons

whose function is both necessary and sufficient for inducing tussling in males" (lines 310-311). The evidence that supports this claim isn't provided. The expression pattern of pC1SS2 neurons in males has been only briefly described in reference 46. It is possible that these neurons overlap with previously characterized dsx+ and/or fru+ neurons that are important for male aggressions (measured by lunges), such as in Koganezawa et al., Curr. Biol. 2016 and Chiu et al., Cell 2020. This adds to the concern that lunge and tussling are not as clearly separated as the authors claim.

While their characterizations of tussling behaviors in wild-type males (Figures 1 and 2) are intriguing, the remaining data have little link with each other, making it difficult to understand what their main conclusion is. Figure 3 suggests that one class of olfactory sensory neurons (OSN) that express Or47b is necessary for tussling behavior. While the authors acknowledged that Or47b-expressing OSNs promote male courtship toward females presumably by detecting cuticular compounds, they provided little discussion on how a class of OSN can promote two different types of innate behavior. No evidence of a functional or circuitry relationship between the Or47b pathway and the pC1SS2 neurons was provided. It is unclear how these two components are relevant to each other. Lastly, the rationale of the experiment in Figure 5 and the interpretation of the results is confusing. The authors attributed a higher mating success rate of older, socially experienced males over younger, socially isolated males to their tendency to tussle, but tussling cannot happen when one of the two flies is not engaged. If, for instance, a socially isolated 14-day-old male does not engage in tussling as indicated in Figure 2, how can they tussle with a group-housed 14-day-old male? Because aggressive interactions in Figure 5 were not quantified, it is impossible to conclude that tussling plays a role in copulation advantage among pairs as authors argue (lines 282-288).

Despite these weaknesses, it is important to acknowledge the authors' courage to initiate an investigation into a less characterized, high-intensity fighting behavior. Tussling requires the simultaneous engagement of two flies. Even if there is confusion over the distinction between lunges and tussling, the authors' conclusion that socially experienced flies and socially isolated flies employ distinct fighting strategies is convincing. Questions that require more rigorous studies are 1) whether such differences are encoded by separate circuits, and 2) whether the different fighting strategies are causally responsible for gaining ethologically relevant resources among socially experienced flies. Enhanced transparency of behavioral data will help readers understand the impact of this study. Lastly, the manuscript often mentions previous works and results without citing relevant references. For readers to grasp the context of this work, it is important to provide information about methods, reagents, and other key resources.

<https://doi.org/10.7554/eLife.104212.1.sa1>

Author response:

Public Reviews:

Reviewer #1 (Public review):

This work addresses an important question in the field of Drosophila aggression and mating- prior social isolation is known to increase aggression in males by increased lunging, which is suppressed by group housing (GH). However, it is also known that single-housed (SH) males, despite their higher attempts to court females, are less successful. Here, Gao et al., developed a modified aggression assay, to address this issue by recording aggression in Drosophila males for 2 hours, over a virgin female which is immobilized by burying its head in the food. They found that while SH males frequently lunge in this assay, GH males switch to higher intensity but very low-frequency tussling.

Constitutive neuronal silencing and activation experiments implicate cVA sensing Or67d neurons promoting high-frequency lunging, similar to earlier studies, whereas Or47b neurons promote low-frequency but higher intensity tussling. Using optogenetic activation they found that three pairs of pC1 neurons- pC1SS2 increase tussling. While P1a neurons, previously implicated in promoting aggression and courtship, did not increase tussling in optogenetic activation (in the dark), they could promote aggressive tussling in thermogenetic activation carried out in the presence of visible light. It was further suggested, using a further modified aggression assay that GH males use increased tussling and are able to maintain territorial control, providing them mating advantage over SI males and this may partially overcome the effect of aging in GH males.

Strengths:

Using a series of clever neurogenetic and behavioral approaches, subsets of ORNs and pC1 neurons were implicated in promoting tussling behaviors. The authors devised a new paradigm to assay for territory control which appears better than earlier paradigms that used a food cup (Chen et al, 2002), as this new assay is relatively clutter-free, and can be eventually automated using computer vision approaches. The manuscript is generally well-written, and the claims made are largely supported by the data.

Thank you for your precise summary of our study and being very positive on the novelty and significance of the study.

Weaknesses:

I have a few concerns regarding some of the evidence presented and claims made as well as a description of the methodology, which needs to be clarified and extended further.

(1) Typical paradigms for assaying aggression in Drosophila males last for 20-30 minutes in the presence of nutritious food/yeast paste/females or all of these (Chen et al. 2002, Nilsen et al., 2004, Dierick et al. 2007, Dankert et al., 2009, Certel & Kravitz 2012). The paradigm described in Figure 1 A, while important and more amenable for video recording and computational analysis, seems a modification of the assay from Kravitz lab (Chen et al., 2002), which involved using a female over which males fight on a food cup. The modifications include a flat surface with a central food patch and a female with its head buried in the food, (fixed female) and much longer adaptation and recording times respectively (30 minutes, 2 hours), so in that sense, this is not a 'new' paradigm but a modification of an existing paradigm and its description as new should be appropriately toned down. It would also be important to cite these earlier studies appropriately while describing the assay.

We will tone down the description and cite related references.

(2) Lunging is described as a 'low intensity' aggression (line 111 and associated text), however, it is considered a mid to high-intensity aggressive behavior, as compared to other lower-intensity behaviors such as wing flicks, chase, and fencing. Lunging therefore is lower in intensity 'relative' to higher intensity tussling but not in absolute terms and it should be mentioned clearly.

We will textually address this issue.

(3) It is often difficult to distinguish faithfully between boxing and tussling and therefore, these behaviors are often clubbed together as box, tussle by Nielsen et al., 2004 in their

Markov chain analysis as well as a more detailed recent study of male aggression (Simon & Heberlein, 2020). Therefore, authors can either reconsider the description of behavior as 'box, tussle' or consider providing a video representation/computational classifier to distinguish between box and tussle behaviors.

We will textually address this issue.

(4) Simon & Heberlein, 2020 showed that increased boxing & tussling precede the formation of a dominance hierarchy in males, and lunges are used subsequently to maintain this dominant status. This study should be cited and discussed appropriately while introducing the paradigm.

We will cite this paper and discuss on this issue.

(5) It would be helpful to provide more methodological details about the assay, for instance, a video can be helpful showing how the males are introduced in the assay chamber, are they simply dropped to the floor when the film is removed after 30 minutes (Figures 1-2)?

We will provide more methodological details.

(6) The strain of Canton-S (CS) flies used should be mentioned as different strains of CS can have varying levels of aggression, for instance, CS from Martin Heisenberg lab shows very high levels of aggressive lunges. Are the CS lines used in this study isogenized? Are various genetic lines outcrossed into this CS background? In the methods, it is not clear how the white gene levels were controlled for various aggression experiments as it is known to affect aggression (Hoyer et al. 2008).

We will textually address this issue.

(7) How important it is to use a fixed female for the assay to induce tussling? Do these females remain active throughout the assay period of 2.5 hours? Is it possible to use decapitated virgin females for the assay? How will that affect male behaviors?

We will textually address this issue and provide additional videos.

(8) Raster plots in Figure 2 suggest a complete lack of tussling in SH males in the first 60 minutes of the encounter, which is surprising given the longer duration of the assay as compared to earlier studies (Nielsen et al. 2004, Simon & Heberlein, 2020 and others), which are able to pick up tussling in a shorter duration of recording time. Also, the duration for tussling is much longer in this study as compared to shorter tussles shown by earlier studies. Is this due to differences in the paradigm used, strain of flies, or some other factor? While the bar plots in Figure 2D show some tussling in SH males, maybe an analysis of raster plots of various videos can be provided in the main text and included as a supplementary figure to address this.

We will textually address the first question and provide more detailed analysis for the second question.

(9) Neuronal activation experiments suggesting the involvement of pC1SS2 neurons are quite interesting. Further, the role of P1a neurons was demonstrated to be involved in increasing tussling in thermogenetic activation in the presence of light (Figure 4, Supplement 1), which is quite important as the role of vision in optogenetic activation experiments, which required to be carried out in dark, is often not mentioned. However,

in the discussion (lines 309-310) it is mentioned that PC1SS2 neurons are 'necessary and sufficient' for inducing tussling. Given that P1a neurons were shown to be involved in promoting tussling, this statement should be toned down.

We will tone down this statement.

(10) Are Or47b neurons connected to pC1SS2 or P1a neurons?

We conducted pathway analysis in the FlyWire electron microscopy database to investigate the connection between Or47b neurons and pC1 neurons. The results indicate that at least three intermediate neurons are required to establish a connection from Or47b neurons to pC1 neurons. Although the FlyWire database currently only contains neuronal data from female brains, they provide a reference for circuit connect in males. Using the currently available upstream and downstream tracing tools (e.g., retro-/trans-Tango), it is not possible to establish a direct connection between the two. Identifying the intermediate neurons involved in this connection is beyond this study. We will discuss on this concern in our revised manuscript.

(11) The paradigm for territory control is quite interesting and subsequent mating advantage experiments are an important addition to the eventual outcome of the aggressive strategy deployed by the males as per their prior housing conditions. It would be important to comment on the 'fitness outcome' of these encounters. For instance, is there any fitness advantage of using tussling by GH males as compared to lunging by SH males? The authors may consider analyzing the number of eggs laid and eclosed progenies from these encounters to address this.

We will discuss on this concern.

Reviewer #2 (Public review):

Summary:

Gao et al. investigated the change of aggression strategies by the social experience and its biological significance by using Drosophila. Two modes of inter-male aggression in Drosophila are known: lunging, high-frequency but weak mode, and tussling, low-frequency but more vigorous mode. Previous studies have mainly focused on the lunging. In this paper, the authors developed a new behavioral experiment system for observing tussling behavior and found that tussling is enhanced by group rearing while lunging is suppressed. They then searched for neurons involved in the generation of tussling. Although olfactory receptors named Or67d and Or65a have previously been reported to function in the control of lunging, the authors found that these neurons do not function in the execution of tussling, and another olfactory receptor, Or47b, is required for tussling, as shown by the inhibition of neuronal activity and the gene knockdown experiments. Further optogenetic experiments identified a small number of central neurons pC1[SS2] that induce the tussling specifically. In order to further explore the ecological significance of the aggression mode change in group rearing, a new behavioral experiment was performed to examine territorial control and mating competition. Finally, the authors found that differences in the social experience (group vs. solitary rearing) are important in these biologically significant competitions. These results add a new perspective to the study of aggressive behavior in Drosophila. Furthermore, this study proposes an interesting general model in which the social experience-modified behavioral changes play a role in reproductive success.

Strengths:

A behavioral experiment system that allows stable observation of tussling, which could not be easily analyzed due to its low frequency, would be very useful. The experimental setup itself is relatively simple, just the addition of a female to the platform, so it should be applicable to future research. The finding about the relationship between the social experience and the aggression mode change is quite novel. Although the intensity of aggression changes with the social experience was already reported in several papers (Liu et al., 2011, etc.), the fact that the behavioral mode itself changes significantly has rarely been addressed and is extremely interesting. The identification of sensory and central neurons required for the tussling makes appropriate use of the genetic tools and the results are clear. A major strength of the neurobiology in this study is the finding that another group of neurons (Or47b-expressing olfactory neurons and pC1[SS2] neurons), distinct from the group of neurons previously thought to be involved in low-intensity aggression (i.e. lunging), function in the tussling behavior. Further investigation of the detailed circuit analysis is expected to elucidate the neural substrate of the conflict between the two aggression modes.

Thank you for the acknowledgment of the novelty and significance of the study, and your suggestions for improving the manuscript.

Weaknesses:

The experimental systems examining the territory control and the reproductive competition in Figure 5 are novel and have advantages in exploring their biological significance. However, at this stage, the authors' claim is weak since they only show the effects of age and social experience on territorial and mating behaviors, but do not experimentally demonstrate the influence of aggression mode change itself. In the Abstract, the authors state that these findings reveal how social experience shapes fighting strategies to optimize reproductive success. This is the most important perspective of the present study, and it would be necessary to show directly that the change of aggression mode by social experience contributes to reproductive success.

We will either tone down this statement or provide additional analysis.

In addition, a detailed description of the tussling is lacking. For example, the authors state that the tussling is less frequent but more vigorous than lunging, but while experimental data are presented on the frequency, the intensity seems to be subjective. The intensity is certainly clear from the supplementary video, but it would be necessary to evaluate the intensity itself using some index. Another problem is that there is no clear explanation of how to determine the tussling. A detailed method is required for the reproducibility of the experiment.

We will provide more detailed methods and data analysis regarding tussling behavior.

Reviewer #3 (Public review):

*In this manuscript, Gao et al. presented a series of intriguing data that collectively suggest that tussling, a form of high-intensity fighting among male fruit flies (*Drosophila melanogaster*) has a unique function and is controlled by a dedicated neural circuit. Based on the results of behavioral assays, they argue that increased tussling among socially experienced males promotes access to resources. They also concluded that tussling is controlled by a class of olfactory sensory neurons and sexually dimorphic central neurons that are distinct from pathways known to control lunges, a common male-type attack behavior.*

*A major strength of this work is that it is the first attempt to characterize the behavioral function and neural circuit associated with *Drosophila* tussling. Many animal species use both low-intensity and high-intensity tactics to resolve conflicts. High-intensity tactics are mostly reserved for escalated fights, which are relatively rare. Because of this, tussling in the flies, like high-intensity fights in other animal species, has not been systematically investigated. Previous studies on fly aggressive behavior have often used socially isolated, relatively young flies within a short observation duration. Their discovery that 1) older (14-days-old) flies tend to tussle more often than younger (2-days-old) flies, 2) group-reared flies tend to tussle more often than socially isolated flies, and 3) flies tend to tussle at a later stage (mostly ~15 minutes after the onset of fighting), are the result of their creativity to look outside of conventional experimental settings. These new findings are key for quantitatively characterizing this interesting yet under-studied behavior.*

Precisely because their initial approach was creative, it is regrettable that the authors missed the opportunity to effectively integrate preceding studies in their rationale or conclusions, which sometimes led to premature claims. Also, while each experiment contains an intriguing finding, these are poorly related to each other. This obscures the central conclusion of this work. The perceived weaknesses are discussed in detail below.

Thank you for the precise summary of the key findings and novelty of the study, and your insightful suggestions.

Most importantly, the authors' definition of "tussling" is unclear because they did not explain how they quantified lunges and tussling, even though the central focus of the manuscript is behavior. Supplemental movies S1 and S2 appear to include "tussling" bouts in which 2 flies lunge at each other in rapid succession, and supplemental movie S3 appears to include bouts of "holding", in which one fly holds the opponent's wings and shakes vigorously. These cases raise a concern that their behavior classification is arbitrary. Specifically, lunges and tussling should be objectively distinguished because one of their conclusions is that these two actions are controlled by separate neural circuits. It is impossible to evaluate the credibility of their behavioral data without clearly describing a criterion of each behavior.

We will add more details in methods.

*It is also confusing that the authors completely skipped the characterization of the tussling-controlling neurons they claimed to have identified. These neurons (a subset of so-called pC1 neurons labeled by previously described split-GAL4 line pC1SS2) are central to this manuscript, but the only information the authors have provided is its gross morphology in a low-resolution image (Figure 4D, E) and a statement that "only 3 pairs of pC1SS2 neurons whose function is both necessary and sufficient for inducing tussling in males" (lines 310-311). The evidence that supports this claim isn't provided. The expression pattern of pC1SS2 neurons in males has been only briefly described in reference 46. It is possible that these neurons overlap with previously characterized *dsx*+ and/or *fru*+ neurons that are important for male aggressions (measured by lunges), such as in Koganezawa et al., Curr. Biol. 2016 and Chiu et al., Cell 2020. This adds to the concern that lunge and tussling are not as clearly separated as the authors claim.*

Reply: we will perform additional morphological and functional experiments on pC1^{SS2} neurons, e.g., whether they are *fru* or *dsx* positive and comparing them with P1a neurons.

While their characterizations of tussling behaviors in wild-type males (Figures 1 and 2) are intriguing, the remaining data have little link with each other, making it difficult to understand what their main conclusion is. Figure 3 suggests that one class of olfactory

sensory neurons (OSN) that express Or47b is necessary for tussling behavior. While the authors acknowledged that Or47b-expressing OSNs promote male courtship toward females presumably by detecting cuticular compounds, they provided little discussion on how a class of OSN can promote two different types of innate behavior. No evidence of a functional or circuitry relationship between the Or47b pathway and the pC1^{SS2} neurons was provided. It is unclear how these two components are relevant to each other. Lastly, the rationale of the experiment in Figure 5 and the interpretation of the results is confusing. The authors attributed a higher mating success rate of older, socially experienced males over younger, socially isolated males to their tendency to tussle, but tussling cannot happen when one of the two flies is not engaged. If, for instance, a socially isolated 14-day-old male does not engage in tussling as indicated in Figure 2, how can they tussle with a group-housed 14-day-old male? Because aggressive interactions in Figure 5 were not quantified, it is impossible to conclude that tussling plays a role in copulation advantage among pairs as authors argue (lines 282-288).

Regarding why Or47b-expressing OSNs regulate two types of innate behaviors, we will add a discussion in the revised manuscript to explore the possible mechanisms underlying this phenomenon.

Regarding the relationship between Or47b-expressing OSNs and pC1^{SS2} neurons, we conducted pathway connection analyses using the FlyWire database. Although the FlyWire database currently only contains neuronal data from female brains, these findings provide a certain degree of reference. The results indicate that at least three intermediate neurons are required to establish the connection between these two neuronal types. We hope the editor and reviewers would agree with us that identifying these intermediate neurons involved in this connection is beyond this study.

Regarding the rationale and conclusions from the experiments in Figure 5, we acknowledge the difficulty in quantifying tussling and lunging behaviors in these experiments. In the revised manuscript, we will tone down the statements about the relationship between fighting strategies and reproductive success. Additionally, we will provide further behavioral experiments to support the association between these two factors.

Despite these weaknesses, it is important to acknowledge the authors' courage to initiate an investigation into a less characterized, high-intensity fighting behavior. Tussling requires the simultaneous engagement of two flies. Even if there is confusion over the distinction between lunges and tussling, the authors' conclusion that socially experienced flies and socially isolated flies employ distinct fighting strategies is convincing. Questions that require more rigorous studies are 1) whether such differences are encoded by separate circuits, and 2) whether the different fighting strategies are causally responsible for gaining ethologically relevant resources among socially experienced flies. Enhanced transparency of behavioral data will help readers understand the impact of this study. Lastly, the manuscript often mentions previous works and results without citing relevant references. For readers to grasp the context of this work, it is important to provide information about methods, reagents, and other key resources.

We will add more details in methods and cite additional references, we will also perform additional experiment on pC1^{SS2} function.

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